

The Augmented Synthetic Historical Control Method

Introduction

The synthetic control method (SCM) by [Abadie et al. \[2010\]](#) is one of the most popular methods in panel data econometrics for treatment effect estimation, second only to difference-in-differences. The idea of an SCM is very simple: it simply posits that the treated unit’s untreated potential outcomes may be understood as some function/weighted average of a set of donor pool units, or unexposed/untreated entities. Many extensions have been proposed to this end, including singular value decomposition [[Amjad et al., 2018](#), [Costa et al., 2023](#)], penalized regression models [[Abadie and L’Hour, 2021](#)], Bayesian methods [[Fernández-Morales et al., 2025](#), [Kim et al., 2020](#)], and other extensions [[Abadie, 2021](#)].

However, as with difference-in-differences, SCMs very generally rely on the availability of a donor pool with which to compare the evolution of the treated outcome to. This may not always hold in practice, particularly with global events like pandemics, financial crises, or in other instances where data for suitable donor units are either impossible to collect or do not exist in principle. [Chen et al. \[2024\]](#) develop the synthetic historical control method (SHC). Instead of untreated donors (like other cities or schools or states), SHC uses the pre-treatment time series of the treated unit as its own donor pool, comparing an evaluation segment to other time periods in the treated unit’s history by breaking the pre-treatment periods into segments. The basic principle of the method is that the future will look like the past, all else equal. [Chen et al. \[2024\]](#) use quadratic programming to solve for the weights for the donor segments, using a weighted average of them to predict the post-treatment counterfactual, using COVID-19 and Brexit as empirical cases.

However, [Chen et al. \[2024\]](#) use convex weights to estimate the counterfactual. This means that the model’s predictions are restricted only to time segments that we have observed before, and may not be suited for complex time-series data where noise or non-smooth dynamics make a pure convex average inadequate. In such cases, exact pre-treatment fit may be impossible, a challenge that also arises in standard SCM applications [[Abadie et al., 2010](#)]. This is precisely the gap ASHC aims to fill: it allows for controlled extrapolation beyond the convex hull, enabling counterfactual predictions even when a perfect convex fit cannot be achieved.

This paper introduces the Augmented Synthetic Historical Control Method (ASHC), which relaxes the strict convexity requirement by allowing limited extrapolation away from the SHC solution. ASHC adds a quadratic penalty on deviations from the SHC weights, together with a stabilizing penalty in directions of low variance. This balances the desire for close fit with the need for stability, ensuring that extrapolation improves flexibility without introducing uncontrolled overfitting. This makes ASHC particularly useful in settings where pre-treatment fit is unattainable with convex weights alone, such as in noisy or highly non-stationary time-series data.

From a theoretical standpoint, ASHC departs from the traditional linear factor model framework used to justify SCM. In SCM, identification is motivated by assuming that untreated potential outcomes follow a factor structure across units. By contrast, SHC and ASHC operate in a purely time-series context: the donor pool consists of overlapping pre-treatment segments of the treated unit itself. Our assumptions therefore follow the semiparametric regression tradition for time-series data. Specifically, we assume that the latent trend is smooth enough to be well-approximated by a truncated Taylor expansion, that noise is bounded (or sub-Gaussian), and that the donor matrix has bounded operator norm. These conditions, while weaker than a full parametric data-generating process, are sufficient to establish finite-sample deviation bounds, asymptotic consistency, and convergence rates for the ASHC estimator.

This paper makes two main contributions. First, we introduce the ASHC estimator, which extends the synthetic historical control framework of [Chen et al. \[2024\]](#) by permitting limited extrapolation beyond the convex hull of donor segments. The estimator combines three components: a fit term that aligns ASHC weights with the evaluation period, a proximity penalty that discourages large deviations from the SHC weights, and a stabilizing penalty that regularizes low-variance directions. Together, these features balance flexibility and robustness, allowing ASHC to provide feasible and stable solutions even in complex or noisy time-series settings.

Second, we establish a set of theoretical properties that clarify the performance guarantees of ASHC. We begin by deriving a finite-sample error bound, showing that ASHC prediction error decomposes into four interpretable components: the SHC residual fit error, a deviation term controlled by effective regularization strength, a smoothness-induced bias from truncating the Taylor approximation of the latent trend, and an additive noise term. We then prove that ASHC is asymptotically consistent: as the length of the pre-treatment period and the smoothing order increase, the prediction error converges to the noise floor, and under sub-Gaussian noise, it converges to zero in probability. Finally, we establish the convergence rate, demonstrating that under an optimally chosen bandwidth, the prediction error decays at rate $O_p(m^{-H/(2H+3)})$, approaching the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order grows. Together, these results show that ASHC preserves the interpretability of SHC while offering provably tighter error control and stronger asymptotic guarantees.

References

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